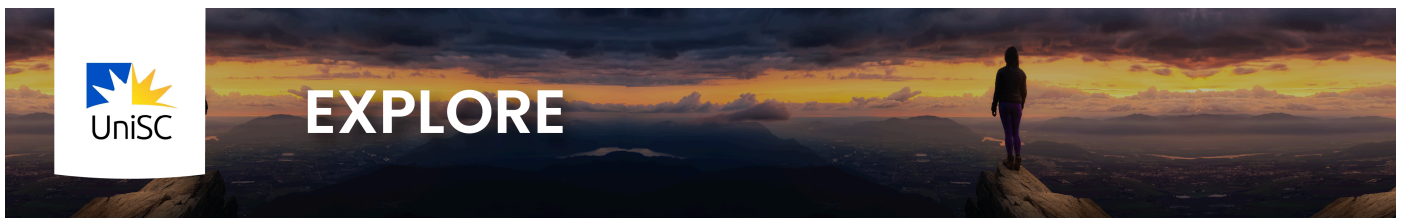


6.3 Risk, Sensitivity, and Scenario Analysis



Guiding Questions

1. What happens if sales are lower than expected?
2. Which variable has the biggest effect on profit or NPV?
3. How does the project perform under best-case and worst-case conditions?
4. What assumptions are most risky?



What you will learn

- Why feasibility analysis must include uncertainty.
- How sensitivity analysis isolates the effect of one variable.
- How scenario analysis combines multiple changes into coherent cases.
- How to identify the most important risk drivers.

A feasibility study should not assume that the base case will happen exactly as forecast. Risk analysis checks how much the project outcome changes when assumptions change.

Sensitivity analysis

Sensitivity analysis changes one variable at a time to see how strongly profit or NPV responds. Common variables include sales volume, selling price, variable cost, fixed cost, and discount rate.

Scenario Analysis

Scenario analysis changes several variables together to create a realistic picture of a possible future. Unlike sensitivity analysis, which changes one variable at a time, scenario analysis reflects the fact that real business conditions rarely change in isolation. A drop in demand, for example, is often accompanied by price pressure and rising costs at the same time.

Common scenarios

- **Base case:** the most likely outcome under current assumptions
 - **Best case:** favourable demand, strong pricing, and controlled costs
 - **Worst case:** weak demand, lower prices, and higher costs
-

Solved Example: The Tutoring Business

We return to the tutoring business introduced earlier. Recall the base case assumptions:

- Price per package: \$200
- Variable cost per package: \$60
- Fixed costs: \$28,000
- Quantity sold: 260 packages

Base case profit:

Contribution margin = $200 - 60 = \$140$ per package

Profit = $(140 \times 260) - 28,000 = 36,400 - 28,000 = \mathbf{\$8,400}$

What if sales fall to 180 packages?

Profit = $(140 \times 180) - 28,000 = 25,200 - 28,000 = \mathbf{-\$2,800}$

The project becomes unprofitable. A drop in quantity alone is enough to push the business into a loss.

Worst case: demand falls and conditions deteriorate further

Now suppose the drop in demand is accompanied by price pressure and rising costs:

- Price per package: \$190
- Variable cost per package: \$70
- Quantity sold: 180 packages

Contribution margin = $190 - 70 = \$120$ per package

Profit = $(120 \times 180) - 28,000 = 21,600 - 28,000 = \mathbf{-\$6,400}$

The loss deepens significantly. This shows that the project is highly exposed to demand risk — and that exposure is compounded when unfavourable conditions occur together, as they often do in practice.

Summary table

Scenario	Price	Variable Cost	Quantity	Profit
Base case	\$200	\$60	260	\$8,400
Demand drop only	\$200	\$60	180	– \$2,800
Worst case	\$190	\$70	180	– \$6,400

The summary table makes it easy to see how quickly profitability deteriorates as conditions worsen. Scenario analysis gives management a clearer picture of the range of possible outcomes and helps identify which combinations of conditions pose the greatest risk to the project's viability.

What sensitivity shows

- It shows the break-even zone and the level of sales needed for profit. It also shows whether the project is fragile or robust. If a small fall in sales causes a large loss, the project carries high operating risk.
- The break-even zone: the approximate number of units sold where profit turns from negative to positive.
- The sales threshold for maintaining a target profit (e.g., 5,000 or 10,000).
- Whether the project is fragile (a small fall in sales causes a large loss) or robust (profit remains positive even under moderate declines).

If a small drop in sales causes a large loss, the project carries high operating risk. Sensitivity analysis helps you quantify that risk and decide whether to:

- Reduce fixed costs.
- Secure minimum-volume contracts.
- Adjust pricing or product mix.
- Build larger safety margins into your assumptions.

Extending the analysis

Beyond the basic one-way table, one can:

- Build multi-variable scenarios as separate blocks in the same worksheet.

- Add conditional formatting to highlight loss-making cases in red.
- Use data tables or goal-seek to back-solve for break-even points.
- Combine sensitivity and scenario analysis into a dashboard that updates automatically when assumptions change.
- Taken together, sensitivity and scenario analysis transform a static feasibility study into a dynamic, risk-aware planning tool that better reflects the real-world uncertainty of business decisions.